

PATRICK LIE

INTERNET OF THINGS / IT (IOT)

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SUMMARY

Fresh graduate BSc (Hons) Information Technology (IoT specialization) from Asia Pacific University (APU). Built end-to-end IoT systems spanning embedded sensing, real-time cloud pipelines, mobile dashboards, alerts, and ML-based insights. Open to Internship / Entry-level roles in Software Engineering, Mobile, IoT, and Cloud.

PROFESSIONAL EXPERIENCE

Alva Production — Software Engineer Intern (App Development) | Remote (WFH) | 2024

- Developed and maintained Flutter mobile apps, contributing to UI screens, reusable components, and feature implementation.
- Supported testing, debugging, and iterative improvements on an existing codebase to improve stability and UX.
- Assisted integration tasks and coordinated progress updates within an ongoing development roadmap.

Allianz — Insurance Agent 2022–2023

- Consulted clients to understand needs and communicate solutions clearly.
- Strengthened problem-solving and stakeholder communication through follow-ups and service support.
- Maintained disciplined tracking of client interactions and deliverables.

PROJECTS

Freshly — Food Freshness Monitoring & Real-Time Spoilage Alerts (FYP, Solo) Completed

Asia Pacific University

Tech: ESP32, Flutter, Firebase (Firestore/FCM), Cloud Functions, Compute Engine, Flask, Scikit-learn

- Built an end-to-end IoT system: ESP32 sensor hub → cloud storage/processing → Flutter dashboard with real-time monitoring and alerts.
- Implemented rule-based notifications (rate-limited) and an ML insight service separated from device/app for maintainability.
- Trained a Random Forest spoilage-risk model using a controlled dataset (~12,000 points) with >99% test accuracy (controlled setup).

DIOT Urban Farming — Smart Monitoring & Automation Pipeline Completed

Asia Pacific University

Tech: MQTT, Node-RED, automation rules, database logging, dashboard visualization

- Designed a scalable IoT pipeline using MQTT for messaging and Node-RED for orchestration and automation.
- Implemented threshold-based monitoring logic, alerting, and data flow design for multi-node deployments.

EDUCATION

Bachelor of Science in Information Technology 2022–2025 (Awaiting graduation)

- **Asia Pacific University (APU)**, Malaysia — BSc (Hons) Information Technology (Internet of Things)

SKILLS

- **Languages:** Python, C, Dart
- **Mobile:** Flutter, Firebase, FCM (push notifications)
- **IoT/Embedded:** ESP32, Arduino IDE, sensor integration, I2C, analog input
- **Integration/Automation:** MQTT, Node-RED
- **Cloud/Backend:** Firebase Firestore, Cloud Functions, Google Compute Engine, Cloud Run, REST APIs
- **Data/ML:** Scikit-learn (Random Forest), Flask, data analysis
- **Web/Tools:** React, TypeScript, Tailwind CSS, Vite, Git, SQL, Cisco Packet Tracer, RStudio, LabVIEW, Figma, Canva

CERTIFICATION

- **IELTS 6.5** (31 Dec 2021)
- **Certificate of Achievement — Alva Productions — Software Engineer Internship (App Department), With Distinction**
- **Cisco Networking Academy — Network Security (Course Completion) — 28 May 2024**

LANGUAGES

- **Indonesian** (Native)
- **English** (IELTS 6.5)
- **Mandarin** (Basic)